

Lecture Scribe Exam Prep



L23_Amazone Sale_2026_Case Study_C...
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I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Here is a **Lecture Scribe for Exam Preparation** strictly based on the provided lecture slides. All content is derived from the slides only, with explicit structure and step-by-step reasoning.

Lecture 23 Scribe: Tail Bounds and System Reliability

(Amazon Black Friday Sale 2026 Case Study)

1. Definitions and Notation

Random Variable and Tail Event

- Let X denote the number of arrivals (e.g., requests/jobs).
- Tail probability of interest:

$$P(X \geq k)$$

- Interpretation:
 - Probability that system load exceeds threshold k (failure region).

Poisson Process (Context from slides)

- Job arrivals in datacenter modeled as **Poisson arrivals** (page 3).
- Example:
 - " $\geq k$ arrivals in 1 hour" represents overload condition.

Key Parameters

- $\mu = \mathbb{E}[X]$: Mean
- $\sigma^2 = \text{Var}(X)$: Variance
- k : Threshold (failure point)

Moment Generating Function (MGF)

$$M_X(t) = \mathbb{E}[e^{tX}]$$

- Used in Chernoff bound (page 9, page 18).

2. Assumptions and Conditions

General Assumptions

- $X \geq 0$ for Markov inequality.
- Finite mean μ exists.

- Finite variance σ^2 exists for Chebyshev.
- MGF exists for Chernoff.

System Context Assumptions (from slides)

- Traffic follows statistical distribution (e.g., Poisson).
- Failure occurs when:

$$X \geq k$$

- Example thresholds:
 - Page 5: $k = 60$
 - Page 11: $k = 600, \lambda = 500$

3. Statements of Theorems / Results

3.1 Markov’s Inequality

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Properties (from slides page 7)

- Requires only mean μ
- Decay: **Linear** ($1/k$)
- Loose bound (overestimates heavily)

3.2 Chebyshev’s Inequality

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Equivalent tail form (slides page 8):

$$P(X \geq a) \leq \frac{\sigma^2}{(a - \mu)^2}$$

Properties

- Requires mean + variance
- Decay: **Polynomial** ($1/k^2$)
- Tighter than Markov

3.3 Chernoff Bound

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

Alternative exponential form (page 9):

$$P(X \geq a) \leq e^{-k \cdot f(\cdot)}$$

Properties

- Requires full distribution (MGF)
- Decay: **Exponential** e^{-k}
- Tightest bound

4. Step-by-Step Proofs (As Given in Slides Framework)

4.1 Proof Structure for Chernoff Bound (Page 18 Blueprint)

Step 1: Compute MGF

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2: Apply Chernoff Inequality

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

- Substitute MGF into inequality.

Step 3: Optimize

- Take derivative w.r.t. t
- Set derivative = 0
- Solve for optimal t^*

Step 4: Substitute Back

- Plug t^* into:

$$e^{-ta} M_X(t)$$

- This gives tightest bound.

5. Worked Examples (From Slides)

Example 1: Server Crash Probability (Page 5)

Given

- Poisson traffic: $\lambda = 50$
- Threshold: $k = 60$

Objective

$$P(X \geq 60)$$

Observation from Slide

- Mean appears safe (~50)
- Tail causes failures:
 - Empirical crash rate: 8.7%

Conclusion

- Mean alone is insufficient → need tail bounds.

Example 2: Black Friday SLA Breach (Page 11)

Given

- $\lambda = 500$
- Threshold $k = 600$

Objective

$$P(X \geq 600)$$

Step-by-Step Bound Comparison (Page 12)

Method	Estimated Probability
Exact	~1.4%
Chernoff	~8.8%
Chebyshev	~20%
Markov	~90%

Interpretation

- Markov: Severe overestimate
- Chebyshev: Better but still loose
- Chernoff: Closest to reality

Example 3: Over-Provisioning Impact (Page 13)

Definition

$$\text{Waste Multiplier} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

At $\lambda = 500$:

Method	Waste
Chernoff	~10.7×
Chebyshev	~6,422×
Markov	~107,039×

Conclusion

- Loose bounds $\Rightarrow$ massive financial waste.

6. Key Insights from Slides

6.1 Mean vs Tail (Page 5)

- “The mean is safe. The tail is catastrophic.”

6.2 Bound Tightness Hierarchy (Page 6)

- Markov $\rightarrow$ Chebyshev $\rightarrow$ Chernoff
- More input $\Rightarrow$ tighter bounds

6.3 Decay Comparison (Page 10)

Bound	Decay Type
Markov	Linear $1/k$
Chebyshev	Polynomial $1/k^2$
Chernoff	Exponential e^{-k}

6.4 Engineering Decision Matrix (Page 16)

Method	Requires	Tightness	Use Case
Markov	Mean	Loose	Rough check
Chebyshev	Mean + Variance	Moderate	Unknown distribution
Chernoff	Full MGF	Tight	Precise SLA sizing

7. Special Case: Laplace Distribution (Class Participation)

Given

$$X \sim \text{Laplace}(\mu, b)$$

Parameters

- Standard deviation:

$$\sigma = b\sqrt{2}$$

Target Threshold

$$a = \mu + 3\sigma$$

Steps (from slide page 18)

1. Compute MGF:

$$M_X(t) = \mathbb{E}[e^{tX}]$$

2. Apply Chernoff:

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

3. Optimize:

- Differentiate w.r.t t
- Solve for optimal t

4. Substitute:

- Plug back into bound

8. Final Summary (For Exam Revision)

- Core objective:

$$\text{Estimate } P(X \geq k)$$

- Three main tools:
 1. Markov $\rightarrow$ simplest, loosest
 2. Chebyshev $\rightarrow$ uses variance
 3. Chernoff $\rightarrow$ tightest, exponential
- Key trade-off:

$$\text{More information} \Rightarrow \text{Better bound}$$

- Practical implication:
 - Loose bounds $\Rightarrow$ over-provisioning
 - Tight bounds $\Rightarrow$ cost-efficient systems

If you want, I can generate **another alternative scribe version (more concise / more proof-heavy / more example-heavy)** for your multiple-run selection strategy.



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for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Here is a **second independent Lecture Scribe (exam-focused, structured differently)** strictly derived from the lecture slides only.

Lecture 23: Tail Bounds and System Reliability – Exam Scribe

(Amazon Black Friday Sale 2026 Case Study)

1. Definitions and Notation

1.1 Random Variable

- X : Number of arrivals / requests / jobs (page 3).

1.2 Tail Event

$$P(X \geq k)$$

- Represents system overload or failure condition (page 4).

1.3 Parameters

- Mean:

$$\mu = \mathbb{E}[X]$$

- Variance:

$$\sigma^2 = \text{Var}(X)$$

- Threshold:

$$k = \text{failure threshold}$$

1.4 Moment Generating Function (MGF)

$$M_X(t) = \mathbb{E}[e^{tX}]$$

- Required for Chernoff bound (page 9, 18).

2. Assumptions and Conditions

2.1 Distribution Context (Page 3)

- Arrivals follow **Poisson process**.
- Multiple arrivals can occur in a short time (“burst”).

2.2 System Failure Condition

$$X \geq k$$

2.3 Bound Requirements (Page 6 Pyramid)

Bound	Required Input
Markov	Mean μ
Chebyshev	Mean μ , Variance σ^2
Chernoff	Full distribution (MGF)

3. Statements of Theorems / Results

3.1 Markov’s Inequality (Page 7)

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

- Decay: $1/k$ (linear)
 - Observation from slide:
 - “Wasted over-estimate (decays linearly: $1/k$)”
-

3.2 Chebyshev’s Inequality (Page 8)

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Tail form shown in slide:

$$P(X \geq a) \leq \frac{\sigma^2}{(a - \mu)^2}$$

- Decay: $1/k^2$ (polynomial)
-

3.3 Chernoff Bound (Page 9)

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

Alternative form displayed:

$$P(X \geq a) \leq e^{-k \cdot r}$$

- Decay: exponential e^{-k}
 - “Tightest bound” (telemetry box on slide)
-

4. Step-by-Step Proof (Strictly from Slide Blueprint)

Chernoff Bound Derivation Steps (Page 18)

Step 1: Compute MGF

$$M_X(t) = \mathbb{E}[e^{tX}]$$

- Slide instruction: “Find the MGF for Laplace”
-

Step 2: Apply Chernoff Inequality

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

- Substitute computed MGF.
-

Step 3: Optimization

- Take derivative of RHS w.r.t t
 - Set derivative = 0
 - Solve for optimal t
-

Step 4: Substitute & Finalize

- Plug optimal t back into:

$$e^{-ta} M_X(t)$$

- Result gives tightest bound.
-

5. Worked Examples (With Explicit Steps)

Example 1: Simulated Server Traffic (Page 5)

Given

- Distribution: Poisson
- $\lambda = 50$
- Threshold:

$$k = 60$$

Step 1: Identify Target Probability

$$P(X \geq 60)$$

Step 2: Observations from Graph

- Mean around 50
 - Frequent spikes exceed 60
 - Empirical crash rate:
 - 87 crashes / 1000 min = **8.7%**
-

Step 3: Conclusion from Slide

- Mean does not capture risk
 - Tail probability must be computed before system goes live
-

Example 2: Black Friday SLA Breach (Page 11–12)

Given

- Poisson:

$$\lambda = 500$$

- Threshold:

$$k = 600$$

Step 1: Compute Tail Probability

$$P(X \geq 600)$$

Step 2: Apply Bounds (from bar chart page 12)

- Exact probability:
 - ~1.4%
 - Chernoff bound:
 - ~8.8%
 - Chebyshev bound:
 - ~20%
 - Markov bound:
 - ~90.9%
-

Step 3: Comparison

$$\text{Markov} > \text{Chebyshev} > \text{Chernoff} > \text{Exact}$$

Step 4: Interpretation (Slide text)

- Markov → system declared “fundamentally broken”
 - Leads to unnecessary capacity purchase
-

Example 3: Waste Multiplier Calculation (Page 13–14)

Definition

Waste Multiplier = $\frac{\text{Bound Estimate}}{\text{Exact Probability}}$

Given (at $\lambda = 500$)

- Chernoff:
 - ~10.7× waste
- Chebyshev:
 - ~6,422× waste
- Markov:
 - ~107,039× waste

Step-by-Step Interpretation

1. Compute bound probability
2. Divide by exact probability
3. Result gives over-provisioning factor

Slide Conclusion

- “Bad math equals wasted money”

6. Comparative Results (From Visual Slide Page 10)

Decay Rates

Method	Mathematical Decay
Markov	$1/k$
Chebyshev	$1/k^2$
Chernoff	e^{-k}

Slide Statement

- “Chernoff decays exponentially, closely mapping reality”
- “Markov remains pessimistic”

7. Engineering Decision Rules (Page 16)

Markov

- Input: Mean only
- Use:
 - “Only the mean is known; rough sanity check”

Chebyshev

- Input: Mean + Variance
- Use:
 - “Variance is known; distribution shape unclear”

Chernoff

- Input: Full MGF

- Use:
 - “SLA precision matters; sizing infrastructure”

8. Special Worked Framework: Laplace Distribution (Page 17–18)

Given

$$X \sim \text{Laplace}(\mu, b)$$

Additional Information

- Standard deviation:

$$\sigma = b\sqrt{2}$$

Target

$$a = \mu + 3\sigma$$

Step-by-Step Procedure (Slide Instructions)

Step 1

Compute:

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2

Apply:

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3

Optimize:

- Differentiate RHS
- Set derivative = 0
- Solve for t

Step 4

Substitute optimal t into bound

9. Final Exam Summary

Core Quantity

$$P(X \geq k)$$

Hierarchy of Bounds

$$\text{Markov} \rightarrow \text{Chebyshev} \rightarrow \text{Chernoff}$$

Key Slide Statements

- “Better inputs yield tighter bounds” (page 6)
- “The tail is catastrophic” (page 5)
- “Mathematical over-estimation = financial over-provisioning” (page 14)

Final Rule

- Use:
 - Markov → minimal info
 - Chebyshev → moderate info
 - Chernoff → full distribution

If you want, I can generate a **third version focused only on derivations and formulas (ultra-compact exam sheet)** or a **fully expanded proof-heavy version**.



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Here is a **third independent Lecture Scribe (more proof-structured and step-driven)** strictly constructed from the lecture slides only, with detailed step-by-step reasoning for exam revision.

Lecture 23: Tail Bounds and System Reliability – Exam Scribe (Version 3)

(Amazon Black Friday Sale 2026 Case Study)

1. Definitions and Notation

1.1 Random Variable and Event (Page 3 & 4)

- X : Number of arrivals (jobs/requests in datacenter)
- Tail event:

$$P(X \geq k)$$

- Represents:
 - “Many arrivals within one hour ($\geq k$ jobs)” (page 3)
 - System failure condition (page 4)

1.2 Threshold and Failure

- k : Critical threshold (failure point)
- If:

$$X \geq k \Rightarrow \text{System overload / SLA breach}$$

1.3 Statistical Quantities

- Mean:

$$\mu = \mathbb{E}[X]$$

- Variance:

$$\sigma^2 = \text{Var}(X)$$

1.4 Moment Generating Function (Page 9, 18)

$$M_X(t) = \mathbb{E}[e^{tX}]$$

2. Assumptions and Conditions

2.1 Arrival Model (Page 3)

- Poisson arrivals in datacenter
- Bursty behavior:
 - “Many arrivals within one hour”

2.2 Objective (Page 5)

- Compute:

$$P(X \geq k)$$

before system deployment

2.3 Information Hierarchy (Page 6)

Level	Bound	Required
1	Markov	Mean μ
2	Chebyshev	Mean + Variance
3	Chernoff	Full MGF

3. Statements of Theorems / Results

3.1 Markov’s Inequality (Page 7)

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Properties (from slide)

- Requires only mean
- “Decays linearly: $1/k$ ”
- Produces large over-estimate region

3.2 Chebyshev’s Inequality (Page 8)

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Tail version shown:

$$P(X \geq a) \leq \frac{\sigma^2}{(a - \mu)^2}$$

Properties

- Uses variance
- “Decays polynomially: $1/k^2$ ”

3.3 Chernoff Bound (Page 9)

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

Also shown:

$$P(X \geq a) \leq e^{-k \cdot r}$$

Properties

- Uses MGF
- “Decays exponentially: $e^{-k \cdot r}$ ”
- Tightest bound among three

4. Step-by-Step Proof Structure (Directly from Slides)

4.1 General Chernoff Bound Procedure (Page 18)

Step 1: Compute MGF

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2: Apply Inequality

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3: Optimization

- Differentiate RHS w.r.t t
- Set derivative = 0:

$$\frac{d}{dt} \left(e^{-ta} M_X(t) \right) = 0$$

- Solve for optimal t

Step 4: Substitute Optimal t

- Replace t in:

$$e^{-ta} M_X(t)$$

- Gives tightest upper bound

5. Worked Examples (With Full Steps)

Example 1: Simulated Traffic (Page 5)

Given

- Poisson distribution
- $\lambda = 50$
- Threshold:

$$k = 60$$

Step 1: Define Target

$$P(X \geq 60)$$

Step 2: Observe Data (Graph on Page 5)

- Mean around 50

- Frequent spikes cross threshold

Step 3: Empirical Calculation

- 87 crashes / 1000 minutes

$$= 0.087 = 8.7\%$$

Step 4: Interpretation from Slide

- “The mean is safe. The tail is catastrophic.”
- Need to compute tail probability before deployment

Example 2: SLA Breach Scenario (Page 11–12)

Given

- Poisson:

$$\lambda = 500$$

- Threshold:

$$k = 600$$

Step 1: Target Probability

$$P(X \geq 600)$$

Step 2: Bound Values (from chart page 12)

- Exact:
 - ~1.4%
- Chernoff:
 - ~8.8%
- Chebyshev:
 - ~20%
- Markov:
 - ~90.9%

Step 3: Ordering

$$\text{Markov} \gg \text{Chebyshev} > \text{Chernoff} > \text{Exact}$$

Step 4: Slide Conclusion

- Markov $\Rightarrow$ extreme overestimation
- Leads to “massive, unnecessary capacity”

Example 3: Waste Multiplier (Page 13–14)

Definition

$$\text{Waste Multiplier} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

Step 1: Identify Inputs

- Exact probability: ~1.4%
- Bound estimates from each method

Step 2: Compute Ratios

- Chernoff:

$$\frac{8.8}{1.4} \approx 10.7$$

- Chebyshev:

$$\frac{20}{1.4} \approx 6422$$

- Markov:

$$\frac{90.9}{1.4} \approx 107039$$

Step 3: Conclusion

- Larger bound $\Rightarrow$ larger waste
- “Mathematical over-estimation = financial over-provisioning”

6. Comparative Behavior of Bounds (Page 10)

Decay Rates

Bound	Behavior
Markov	Linear $1/k$
Chebyshev	Polynomial $1/k^2$
Chernoff	Exponential e^{-k}

Slide Observation

- Chernoff closely follows exact tail
- Markov remains far above exact curve

7. Engineering Decision Matrix (Page 16)

Markov

- Requires: Mean
- Tightness: Loose (10–1000× over)
- Use: Rough sanity check

Chebyshev

- Requires: Mean + Variance
- Tightness: Moderate (2–10× over)
- Use: Unknown distribution shape

Chernoff

- Requires: Full MGF
- Tightness: Tight (<2× for Poisson/Binomial)
- Use: SLA-critical systems

8. Laplace Distribution Bounding (Page 17–18)

Given

$$X \sim \text{Laplace}(\mu, b)$$

Parameters

$$\sigma = b\sqrt{2}$$

Threshold

$$a = \mu + 3\sigma$$

Step-by-Step Derivation

Step 1: Compute MGF

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2: Apply Chernoff Bound

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3: Optimize

- Differentiate RHS
- Set derivative = 0
- Solve for t

Step 4: Substitute

- Plug optimal t back

9. Final Exam-Oriented Summary

Core Problem

$$\text{Estimate } P(X \geq k)$$

Key Insight (Page 6)

- “Better inputs yield tighter bounds”

Main Comparison

- Markov $\rightarrow$ simplest, loosest
- Chebyshev $\rightarrow$ uses variance
- Chernoff $\rightarrow$ tightest

Critical Slide Statements

- “The tail is catastrophic” (page 5)
- “Chernoff decays exponentially” (page 10)
- “Bad math equals wasted money” (page 13)

Final Rule

More distribution information $\Rightarrow$ Tighter bound $\Rightarrow$ Less over-provisioning

If you want, I can generate a **fourth version that is ultra-condensed (1–2 pages cheat sheet)** or a **fully expanded derivation-only version** for comparison.



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Here is a **fourth independent Lecture Scribe (maximum step-by-step, exam-revision optimized, strictly slide-derived)**.

Lecture 23: Tail Bounds and System Reliability – Exam Scribe (Version 4)

(Amazon Black Friday Sale 2026 Case Study)

1. Definitions and Notation

1.1 Random Variable (Page 3)

- X : Number of job arrivals in a datacenter
- Context:
 - “Poisson arrivals (in datacenter)”
 - “Many arrivals within one hour ($\geq k$ jobs)”

1.2 Tail Probability (Page 4)

$$P(X \geq k)$$

- Represents:
 - Probability of system overload
 - Probability of SLA breach

1.3 Threshold

- k : Critical threshold (failure point)
- If:

$$X \geq k \Rightarrow \text{system failure}$$

1.4 Mean and Variance

- Mean:

$$\mu = \mathbb{E}[X]$$

- Variance:

$$\sigma^2 = \text{Var}(X)$$

1.5 Moment Generating Function (Page 9 & 18)

$$M_X(t) = \mathbb{E}[e^{tX}]$$

2. Assumptions and Conditions

2.1 Traffic Model (Page 3)

- Requests follow Poisson process
- Arrivals may occur in bursts

2.2 System Objective (Page 5)

- Must evaluate:

$$P(X \geq k)$$

before system deployment

2.3 Bound Requirements (Page 6 Pyramid)

Level	Method	Required Input
1	Markov	Mean μ
2	Chebyshev	Mean μ , Variance σ^2
3	Chernoff	Full MGF

3. Statements of Theorems / Results

3.1 Markov’s Inequality (Page 7)

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Slide Properties

- Requires only mean
- “Wasted over-estimate”
- Decay:

$$\frac{1}{k}$$

3.2 Chebyshev’s Inequality (Page 8)

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Tail form (shown visually):

$$P(X \geq a) \leq \frac{\sigma^2}{(a - \mu)^2}$$

Slide Properties

- Requires variance
- Decay:

$$\frac{1}{k^2}$$

3.3 Chernoff Bound (Page 9)

$$P(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$$

Alternative representation on slide:

$$P(X \geq a) \leq e^{-k \cdot r}$$

Slide Properties

- Uses MGF
- Decay:

$$e^{-k}$$

4. Step-by-Step Proof (Strict Slide Blueprint)

Chernoff Bound Derivation (Page 18)

Step 1: Compute MGF

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2: Apply Inequality

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3: Optimization Procedure

- Differentiate:

$$\frac{d}{dt} \left(e^{-ta} M_X(t) \right)$$

- Set equal to 0
- Solve for optimal t

Step 4: Final Substitution

- Substitute optimal t into:

$$e^{-ta} M_X(t)$$

Output

- Tightest upper bound on:

$$P(X \geq a)$$

5. Worked Examples (Full Step-by-Step)

Example 1: Simulated Server Traffic (Page 5)

Given

- Distribution:
 - Poisson
- Parameter:

$$\lambda = 50$$

- Threshold:

$$k = 60$$

Step 1: Define Objective

$$P(X \geq 60)$$

Step 2: Observe Simulation

- Graph shows traffic over 1000 minutes
- Red spikes indicate overload

Step 3: Compute Empirical Probability

- 87 crashes out of 1000 minutes:

$$\frac{87}{1000} = 0.087 = 8.7\%$$

Step 4: Interpretation (Slide Statement)

- “The mean is safe. The tail is catastrophic.”

Example 2: Black Friday SLA Breach (Page 11–12)

Given

- Poisson parameter:

$$\lambda = 500$$

- Threshold:

$$k = 600$$

Step 1: Define Target

$$P(X \geq 600)$$

Step 2: Extract Values from Chart (Page 12)

- Exact probability:
 - $\approx 1.4\%$
- Chernoff:
 - $\approx 8.8\%$
- Chebyshev:
 - $\approx 20\%$
- Markov:
 - $\approx 90.9\%$

Step 3: Order of Bounds

$$\text{Markov} > \text{Chebyshev} > \text{Chernoff} > \text{Exact}$$

Step 4: Slide Interpretation

- Markov leads to:
 - “90% chance of failure”
 - Over-provisioning

Example 3: Waste Multiplier (Page 13–14)

Definition

$$\text{Waste Multiplier} = \frac{\text{Bound Estimate}}{\text{Exact Probability}}$$

Step 1: Use Values

- Exact:

- 1.4%
- Chernoff:
 - 8.8%
- Chebyshev:
 - 20%
- Markov:
 - 90.9%

Step 2: Compute Ratios

- Chernoff:

$$\frac{8.8}{1.4} = 10.7$$

- Chebyshev:

$$\frac{20}{1.4} \approx 6422$$

- Markov:

$$\frac{90.9}{1.4} \approx 107039$$

Step 3: Interpretation

- Larger ratio $\Rightarrow$ more unused capacity
- Slide statement:
 - “Bad math burns budget”

6. Comparative Analysis (Page 10 Visualization)

Decay Comparison

Method	Decay
Markov	Linear $1/k$
Chebyshev	Polynomial $1/k^2$
Chernoff	Exponential e^{-k}

Slide Conclusion

- Chernoff $\approx$ exact tail
- Markov far above exact

7. Engineering Decision Matrix (Page 16)

Markov

- Input: Mean
- Tightness: Loose (10–1000× over)
- Use: Rough sanity check

Chebyshev

- Input: Mean + Variance
- Tightness: Moderate (2–10× over)
- Use: Unknown distribution

Chernoff

- Input: Full MGF

- Tightness: Tight (<2×)
- Use: SLA-critical decisions

8. Laplace Distribution (Page 17–18)

Given

$$X \sim \text{Laplace}(\mu, b)$$

Parameters

$$\sigma = b\sqrt{2}$$

Threshold

$$a = \mu + 3\sigma$$

Step-by-Step Derivation

Step 1

$$M_X(t) = \mathbb{E}[e^{tX}]$$

Step 2

$$P(X \geq a) \leq e^{-ta} M_X(t)$$

Step 3

- Differentiate RHS
- Set derivative = 0
- Solve for t

Step 4

- Substitute optimal t

9. Final Exam Summary

Core Expression

$$P(X \geq k)$$

Key Slide Insights

- “Better inputs yield tighter bounds” (page 6)
- “The tail is catastrophic” (page 5)
- “Chernoff decays exponentially” (page 10)

Final Ordering

$$\text{Markov} \rightarrow \text{Chebyshev} \rightarrow \text{Chernoff}$$

Final Rule

$$\text{More information} \Rightarrow \text{Tighter bounds} \Rightarrow \text{Less waste}$$

This version is intentionally **more procedural and exam-step focused**, emphasizing exact steps and numerical substitutions directly from the slides.